

2019

① a) $26 \cdot 26 \cdot 26 = 17576$

b) # ways to choose a pair where

i) order does not matter ($AK = KA$)

$$\binom{26}{2} = 325$$

ii) first wire: $\binom{26}{2}$

second wire: $\binom{24}{2}$

wires are the same $\rightarrow$

1: $A \rightarrow B$

2: $C \rightarrow D$

2: $A \rightarrow B$

1: $C \rightarrow D$

Thus,

$$\frac{\binom{26}{2} \binom{24}{2}}{2} = 44850$$

2) a) $P(\text{Gang} | 1950)$

$$= \frac{P(\text{Gang} \cap 1950)}{P(1950)} \leftarrow \frac{|\text{Gang} \cap 1950|}{\cancel{11950}} \leftarrow \frac{151}{151} \leftarrow \text{total population}$$

$$= \frac{|\text{Gang} \cap 1950|}{151}$$

$$= \frac{|\text{Gang} \cap 1950|}{151} = \frac{|\text{Gang} \cap 1950|}{11950}$$

$$\boxed{\frac{P(\text{Gang} \cap 1950)}{P(1950)}} = \frac{P(\text{Gang} \cap 1950)}{11950}$$

$$\frac{P(\text{Gang} \cap 1950)}{P(1950)} = \frac{P(\text{Gang} \cap 1950)}{11950}$$

$$b) P(1950 | c, G_{any})$$

$$= \frac{P(G_{any}, c | 1950) P(1950)}{P(G_{any}, c)}$$

Bayes

$$P(Y | c, G) = \frac{P(Y, c, G)}{P(c, G)}$$

$$P(G | c) = \frac{P(G, c)}{P(c)} \Rightarrow P(G, c) = P(G | c) \cdot P(c)$$

$$P(G | Y, c) = \frac{P(G, Y, c)}{P(Y, c)} \Rightarrow P(G, Y, c) = P(G | Y, c) \cdot P(Y, c)$$

$$P(Y | c, G) = \frac{P(G | Y, c) P(Y, c)}{P(G | c) \cdot P(c)}$$

$$= \frac{P(G | Y, c) \cdot P(Y | c)}{P(G | c)}$$

$$\frac{P(G|Y, C) \cdot P(Y|C)}{P(G|C)}$$

$$= \frac{P(G|Y, C) \cdot P(Y|C)}{\sum_{y=1901}^{2019} P(G|y, C) P(y|C)} \quad \text{LOTP}$$

$$= \frac{P(G|Y) \cdot P(Y|C)}{\sum_{y=1901}^{2019} P(G|y) \cdot P(y|C)} \quad \text{using assumption}$$

$$= \frac{[(a)] \cdot \text{prior}(1950)}{\sum_{y=1901}^{2019} \left(\frac{\text{count}(y, G)}{\sum_{i=1}^k \text{count}(y, N_i)} \right) \cdot \text{prior}(y)} \quad \text{follow (a)}$$

2017

① multinomial coefficient or

a) bucketing into fixed sized containers

$$\binom{12}{5,4,3} = \frac{12!}{5!4!3!}$$

① b) $\rightarrow$ select 2 fruits of same type to pair
(only 1 way to do this for each fruit)
 $\rightarrow$ remaining 10 fruits are distributed to
" " students

apples to pair OR mandarin to pair OR peaches to pair

$$\binom{10}{3,4,3} + \binom{10}{5,2,3} + \binom{10}{5,4,1}$$

① c) possible scenarios: 2 apples left

OR

$$\binom{10}{3,4,3} + \binom{10}{5,2,3} + \binom{10}{5,4,1}$$

2 mand
2 peach

$$\left. \begin{array}{l} 1 a 1 m \\ 1 a 1 p \\ 1 m 1 p \end{array} \right\}$$
$$+ \binom{10}{4,3,3} + \binom{10}{4,4,2} + \binom{10}{5,3,2}$$

2017

③ a)

We are given PMF
plug in $i = 10$

$$P(X=10) = \frac{\frac{1}{10}}{\sum_{h=1}^{30} \frac{1}{n}} \approx 17.8$$

2018

① a) $26 \cdot 26 \cdot 26 \cdot 26$

$$b) P(\text{English}) = \frac{5500}{26^4}$$

$$c) P(\text{at least one is English}) =$$

from ~~the~~ our generations

$$1 - P(\text{none are English})$$

from our generations

$$\frac{|E|}{|S|} \leftarrow \begin{array}{l} \# \text{ ways to choose generations} \\ \text{that are not English} \end{array}$$
$$\leftarrow \begin{array}{l} \# \text{ ways to choose} \\ \text{our generations} \end{array}$$

$$|S| = \binom{26^4}{100,000}$$

$$|E| = \binom{26^4 - 5500}{100,000}$$

$$\textcircled{2} a) P(D=1) = \sum_{i,j \in \{0,1\}} P(D=1, G=i, T=j) \quad \underline{\text{LOTP}}$$

$$= 0.01 + 0.01 + 0.02 + 0.08 = 0.12$$

$$b) P(G=1|D=1) = \frac{P(G=1, D=1)}{P(D=1)} \quad \underline{\text{Bayes}}$$

$$= \frac{P(G=1, D=1, T=0) + P(G=1, D=1, T=1)}{P(D=1)}$$

$$= \frac{0.01 + 0.08}{0.12} = 0.75$$

Parity:

$$c) \text{Parity: } P(G=1|D=0) = P(G=1|D=1)$$

$$\frac{0.32 + 0.28}{0.88}$$

$$\approx 0.68$$

$$\downarrow$$

$$\cancel{0.32 + 0.28}$$

$$\cancel{0.01 + 0.08}$$

$$0.75$$

$$d) \text{calibration: } P(G=T|D=0) = P(G=T|D=1)$$

$$P(G=1, T=1|D=0) + P(G=0, T=0|D=0)$$

$$\frac{0.28 + 0.21}{0.88} \approx 0.56$$

$$P(G=1, T=1|D=1) + P(G=0, T=0|D=1)$$

$$\frac{0.08 + 0.01}{0.12} \approx 0.75$$

2018

$$e) P(G=1 | D=0, T=1) = P(G=1 | D=1, T=1)$$

$$P(G=1 | D=1, T=1) = \frac{P(G=1, D=1, T=1)}{P(D=1, T=1)}$$

$$= \frac{0.08}{0.08 + 0.02} = 0.8$$

$$P(G=1 | D=0, T=1) = \frac{P(G=1, D=0, T=1)}{P(D=0, T=1)}$$

$$= \frac{0.28}{0.28 + 0.07} = 0.8$$

$$5a) P(A) = 0.8$$

$$P(B) = 0.2$$

$$P(E|A) = 0.6 \quad P(E^c|A) = 0.4$$

$$P(E|B) = 0.2 \quad P(E^c|B) = 0.8$$

$$P(A|E^c) = \frac{P(A, E^c)}{P(E^c)}$$

Chain Rule: $P(A, B) = P(A|B) \cdot P(B)$

$$= \frac{P(E^c|A)P(A)}{P(E^c|B)P(B) + P(E^c|A)P(A)} = \frac{0.4 \cdot 0.8}{0.8 \cdot 0.2 + 0.4 \cdot 0.8} = \frac{2}{3}$$